Tran Ta Quang Minh

Google Scholar | ttqminh2005@gmail.com | LinkedIn: ttqminh | GitHub: coutMinh | *(September 2026)*

Education

- **University of Science, Vietnam National University Ho Chi Minh City (VNU-HCM)**
  Bachelor of Science in Artificial Intelligence Sept 2023 – July 2027
  GPA: 9.41/10 (Rank 2/56) – Excellent Student for 6 consecutive semesters

Research & Academia Experience

- **Trustworthy AI & Robustness** *August 2025 – Present*
  Undergraduate Researcher Advisors: Assoc. Prof. Suhang Wang, Tung Kieu; Ph.D. Student Cuong Dang Cao
  - **Uncertainty Quantification for RAG (URAG)** : Proposed a comprehensive benchmark evaluating both predictive accuracy and uncertainty across multiple retrieval-augmented methods, covering 8 datasets over 5 domains. Developed a standardized pipeline converting open-ended RAG tasks into a multiple-choice format, enabling quantitative uncertainty assessment via conformal prediction. *Under review NeurIPS'26*
  - **Retrieved But Not Reliable: A Survey on Attacks, and Defenses in Retrieval-Augmented Generation** : Conducted a pipeline-aware survey of Retrieval-Augmented Generation (RAG) robustness and security, systematically organizing threat models, attacks, defenses, benchmarks, and explainability methods across the corpus, retriever, reranker, generator, and traceback stages. *Accepted to Findings of EMNLP'26*
  - **When Unlearning Hurts: Adversarial Forgetting in Graph Neural Networks** : Investigating adversarial attacks that exploit machine-unlearning procedures to induce persistent degradation across sequential tasks in graph neural networks. *On going project*.

- **Linguistics** *May 2026 – Present*
  Undergraduate Researcher Advisor: Duc Hoang, MIT Ph.D.
  - **How Robust Are LLMs to Vietnamese Dialects?** : Introduced *VialectBench*, a controlled benchmark with 400 Standard Vietnamese source instances and 2,400 human-written dialectal rewrites across six dialect groups and four language-understanding tasks. Evaluated instruction-tuned LLMs and quantified dialect-induced performance degradation, prediction inconsistency, and harmful answer flips. *Under review AAAI'27*.

- **Video Editing & Generation** *May 2026 – Present*
  Undergraduate Researcher Advisor: Thanh Le, Ph.D.
  - **Adversarial Attack on Video Editing models**: *On going project*.

Publications

[1] URAG: A Benchmark for Uncertainty Quantification in Retrieval-Augmented Large Language Models *(Under review at NeurIPS 2026)*
Vinh Nguyen*, Cuong Dang*, Jiahao Zhang, Hoa Tran, **Minh Tran**, Trinh Chau, Thai Le, Lu Cheng, Suhang Wang.

[2] Retrieved But Not Reliable: A Survey on Attacks, and Defenses in Retrieval-Augmented Generation *(Findings of EMNLP 2026)*
Minh Tran, Cuong Dang, Tuc Nguyen, Khanh-Tung Tran, Minh Huynh Nguyen, Trinh Chau, Kien Le, Do Xuan Long, Jiahao Zhang, Fali Wang, Hoang D. Nguyen, Thanh Le, Suhang Wang.

[3] How Robust Are LLMs to Vietnamese Dialects? *(Under review at AAAI 2027)*
Minh Tran, Trinh Chau, Thanh-Nhan Le, Nam Tran, Luan Thanh Nguyen, Cuong Dang, Duc Hoang.

Projects & Awards

- Recognized among the top 3% of students in the Faculty of Information Technology (2023–2024).
- Recognized among the top 3% of students in the Faculty of Information Technology (2024–2025).
- Received a school scholarship for the second semester of the 2023–2024 academic year.
- Honored to be selected as one of 50/200 research students to participate in the Agentic AI summer camp organized by HUST, and received the First Runner-Up Award for the project presentation.
- **AI Assistant Glasses** : Winner of HCMUSxBosch AI Innovation Competition 2024. Built smart glasses that fuse audio & visual data for danger alerts to assist the visually impaired; implemented an LSTM to predict future states of moving objects.

- **EVENTA - Event-based Image Retrieval** : Second Prize - Engineered a two-phase retrieval system: offline OpenCLIP (ViT-bigG-14) embeddings + semantic chunking; online fusion of dense (Nomic-v1.5) & sparse (BM25) via RRF with multi-stage reranking; used PyTorch, LangChain, Transformers. My team also had the opportunity to present method at ACMMM workshop.
- **AquaSafe** : Honor Prize - Data4Life Hack for Growth 2025.
- **Real-time Monitoring Surveillance System** : SOICT Hackathon 2024 Finalist.
- **Lifelog Retrieval App** : AI Challenge HCMC 2024 Finalist.
- **Flood Forecasting Model** : Developed a model to forecast flooding up to seven days in advance, tailored to Vietnamese weather conditions. Mentored the project as part of the SEAS Summer School.
- **Local Fish Identification** : An edge-optimized machine learning system for real-time identification of fish species native to Dong Hoi, Quang Binh (Central Vietnam). Mentored the project as part of the SEAS Summer School.

Teaching & Mentorship

- **Summer in Engineering and Applied Sciences (SEAS)** *June 2025 – Present*
  Teaching Assistant Supervisor: Duc Hoang, MIT Ph.D.
  - **Bootcamp Instruction**: Assisted a two-week intensive program for 40 top high school students in Central Vietnam covering Python, Linear Algebra, Machine Learning, and Applied AI (Computer Vision, AI for Science, Network AI). Worked in an intensive academic environment alongside PhDs from Harvard, MIT, Stony Brook, and NUS to help develop the teaching program.
  - **Exercise Development**: Designed comprehensive Linear Algebra Python exercises.
  - **Applied Project**: Mentored the design of two Vietnam-centric AI applications: (1) a flood forecasting model using filtered U.S. CAMELS hydrologic data and Open-Meteo inputs to predict discharge in the Long Dai River, enabling comparison with Google Flood Hub; (2) a computer vision system with FishNet dataset and edge-deployment techniques for real-time marine fish species classification on resource-constrained hardware.
- **Science Lab** *2025 – 2026*
  Mentor & Competition Judge
  - **Summer Hackathon: Software Engineering 2025** : Mentored a student team during the Science Lab Summer Hackathon in Software Engineering. Guided students through project development and technical problem solving, and was recognized by the organizers for excellent mentorship.
  - **AIHACKS – Datathon 2026** : Served as a judge for a digit-classification Datathon, evaluating solutions from multiple high school teams using diverse machine-learning approaches based on methodology, model performance, problem-solving strategy, and presentation.

Technical Skills

- **Languages**: Python (scikit-learn, NumPy, Pandas, PyTorch, ...), C++.
- **Research Areas**: NLP, Robustness & Trustworthy AI, Information Retrieval, Video Editing.
- **Soft Skills**: Leadership, Teamwork, Presentation Skills.